

PETALNEX PVT LTD

Internship — Daily Progress Report

Intern Name	Muhammad Bin Waseem
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Daily Report Fields

Today I Learned	I learned the fundamentals of Generative AI and Large Language Models (LLMs), including the differences between AI, Machine Learning, Generative AI, and LLMs. I also learned about tokens, context windows, hallucinations, system prompts, user prompts, API-based model access, and compared providers such as Google Gemini and Groq.
Today I Built / Practiced	I generated API keys for Gemini AI Studio and Groq, tested both providers using Postman, and learned how to send prompts to LLM APIs. I also compared the strengths of each provider for different automation tasks and documented my findings.
Technologies / Tools Used	Google AI Studio, Groq Console, Postman, REST APIs, JSON, HTTP POST Requests, Gemini API, Groq API
Problem(s) Faced	While testing Gemini, the API returned a quota exceeded (429) error because the project had no available free-tier quota. I also encountered a model availability issue when initially using an older model endpoint.
How I Solved Them	I updated the request to use the latest supported model endpoint and verified that the request format was correct. Since the remaining issue was related to API quota rather than the request itself, I documented the limitation and successfully completed the API testing using Groq, which responded correctly.
Task Status	Completed

GitHub / Workflow Link	https://github.com/MuhammadBinWaseemm/AI-Automation-Internship-Pentalnex-Pvt.-Ltd/tree/main/daily-logs/DAY-13
Plan for Next Day	Looking forward with LLMs & their api calls.
Comparison Note	During this practical task, I tested both Google Gemini and Groq APIs using Postman. The Groq API responded successfully, while the Gemini API returned a quota-related error because the free-tier quota for my project was unavailable. Even though I could not complete a successful Gemini request, I compared both providers based on their capabilities. For classification tasks, I would choose Groq. It provides extremely fast inference, making it ideal for automating workflows such as lead classification, email categorization, sentiment analysis, and data processing in n8n. Since these tasks usually involve short prompts and require quick responses, Groq offers excellent speed and low latency. For long-context tasks, I would choose Google Gemini. Gemini models support larger context windows, making them more suitable for summarizing lengthy documents, analyzing reports, answering questions from large datasets, and processing extensive conversations. Although responses may be slower than Groq, the ability to handle much larger amounts of information makes Gemini the better choice for complex reasoning tasks. Overall, Groq is better suited for fast automation and classification, while Gemini is preferable for document understanding and long-context reasoning. The choice depends on whether speed or context capacity is the primary requirement.